

ECO102 - Introduction to Macroeconomics

Department of Economics and Social Sciences (ESS), BRAC University

Economic Growth

Lecture 3

Reading

- Economics by Michael Parkin, 10th edition
 - Chapter: 23

Outline

- ❖ Define and calculate the economic growth rate and explain the implications of sustained growth
- ❖ Explain how population growth and labor productivity growth make potential GDP grow
- ❖ Explain the sources of labor productivity growth
- ❖ Explain the theories of economic growth

Economic Growth

- **Economic growth** is a sustained expansion of production possibilities measured as the increase in real GDP over a given period
- Rapid economic growth maintained over a number of years can transform a poor nation into a rich one
- Slow economic growth or the absence of growth can condemn a nation to devastating poverty

Calculating Growth Rates

We express the economic growth rate as the annual percentage change of real GDP

$$\text{Real GDP growth rate} = \frac{\text{Real GDP in current year} - \text{Real GDP in previous year}}{\text{Real GDP in previous year}} \times 100$$

For example, if real GDP in the current year is \$11 trillion and if real GDP in the previous year was \$10 trillion, then the economic growth rate is 10 percent

The growth rate of real GDP tells us how rapidly the total economy is expanding, but it does not tell us about changes in the standard of living

Calculating Growth Rates

The standard of living depends on real GDP per person (also called per capita real GDP), which is real GDP divided by the population.

	Current Year	Previous Year
Real GDP	\$11 trillion	\$10 trillion
Population	202 million	200 million
Real GDP per person	\$54,455	\$50,000

$$\begin{array}{l} \text{Real GDP} \\ \text{per person} \\ \text{growth rate} \end{array} = \frac{\$54,455 - \$50,000}{\$50,000} \times 100 = 8.9 \text{ percent}$$

Calculating Growth Rates

An alternative way of calculating the growth rate of real GDP per person

- If the growth rate of real GDP is 10 percent and the population changes from 200 million to 202 million, then population growth rate is 1 percent
- Therefore, the growth rate of real GDP per person approximately (10 percent – 1 percent) equals 9 percent
 - Real GDP per person grows only if real GDP grows faster than the population grows
 - If the growth rate of the population exceeds the growth of real GDP, then real GDP per person falls

Economic Growth - Increase in Potential GDP

- Economic growth occurs when real GDP increases
- But a one-shot rise in real GDP or a recovery from recession isn't economic growth
- Economic growth is a sustained, year-after-year increase in potential GDP

What determines Potential GDP?

- Labor, capital, land, and entrepreneurship produce real GDP, and the productivity of the factors of production determines the quantity of real GDP that can be produced
- The quantity of land is fixed and on any given day, the quantities of entrepreneurial ability and capital are also fixed and their productivities are given
- The quantity of labor employed is the only variable factor of production
- **Potential GDP** is the level of real GDP when the quantity of labor employed is the full-employment quantity

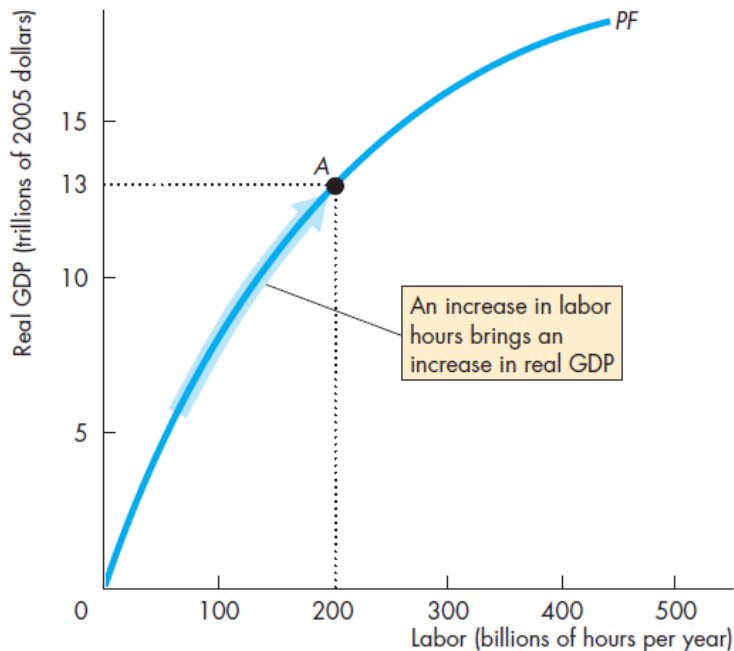
What determines GDP?

To determine potential GDP, we use a model with two components:

- **An aggregate production function**
- **An aggregate labor market**

Aggregate Production Function

FIGURE 23.4 The Aggregate Production Function



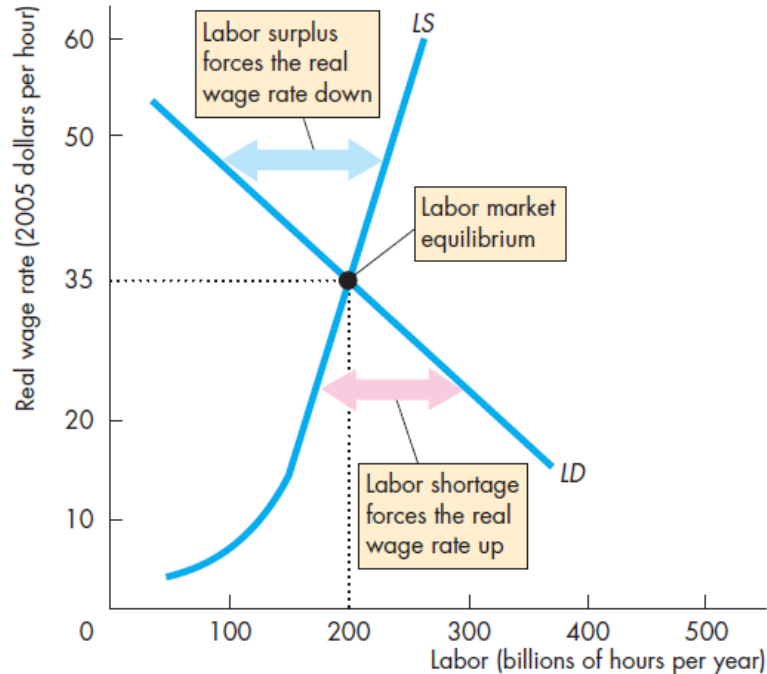
- The aggregate production function is the relationship that tells us how real GDP changes as the quantity of labor changes, when all other influences on production remain the same
- An increase in the quantity of labor (and a corresponding decrease in leisure hours) brings a movement along the production function and an increase in real GDP

Aggregate Labor Market

- The Demand for Labor – The quantity of labor demanded is the number of labor hours hired by all the firms in the economy during a given period and this quantity depends on the price of labor, which is the real wage rate
- The real wage rate – influences the quantity of labor demanded
 - The quantity of labor demanded increases as the real wage rate decreases
- The Supply of Labor – The quantity of labor supplied is the number of labor hours that all the households in the economy plan to work during a given period. This quantity depends on the real wage rate
 - The quantity of labor supplied increases as the real wage rate increases

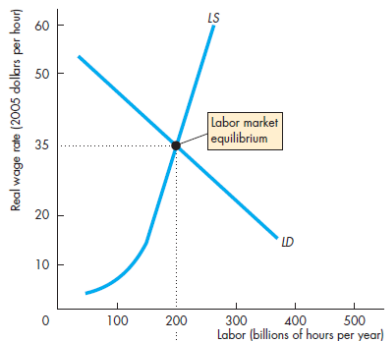
Labor Market Equilibrium

FIGURE 23.5 Labor Market Equilibrium

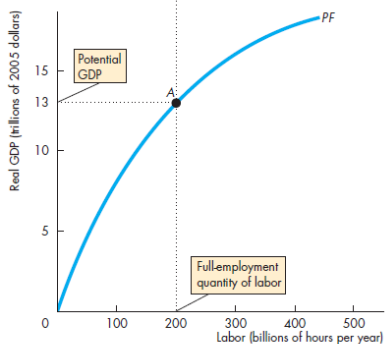


- The forces of supply and demand operate in labor markets to eliminate a shortage or a surplus
- When there is neither a shortage nor a surplus, the labor market is in equilibrium—at this equilibrium real wage rate and level of employment, the economy is at full employment

The Labor Market & Potential GDP



(a) The labor market



(b) Potential GDP

- Part (a) shows labor market equilibrium – at the equilibrium real wage rate, equilibrium employment is 200 billion hours
- Part (b) shows the production function –with 200 billion hours of labor, the economy can produce a real GDP of \$13 trillion – this amount is potential GDP
- Therefore, at the equilibrium quantity of labor, the economy is at full employment, and the quantity of real GDP at full employment is potential GDP

What Makes Potential GDP Grow?

We can divide all the forces that make potential GDP grow into two categories:

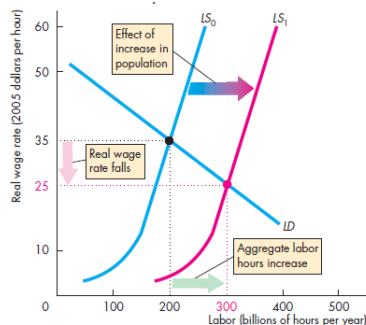
- Growth of the supply of labor
- Growth of labor productivity

Growth of the Supply of Labor

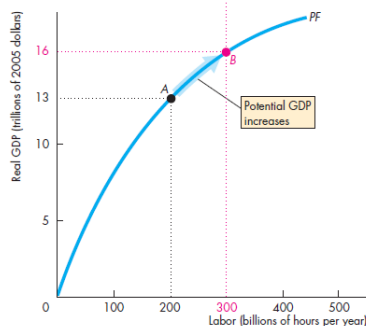
The quantity of labor changes as a result of changes in

1. Average hours per worker
2. The employment-to-population ratio
3. The working-age population

The Effects of an Increase in Population



(a) The labor market

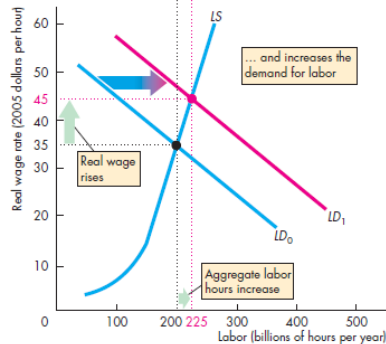
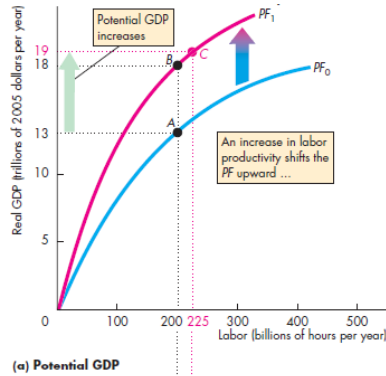


- An increase in the population increases the supply of labor
- With an increase in the supply of labor (supply of labor curve shifts rightward) and no change in the demand for labor, the real wage rate falls and the equilibrium quantity of labor increases (aggregate labor hours increase)
- The increased quantity of labor produces more output and potential GDP increases
- However, **diminishing returns** bring a decrease in potential GDP per hour of labor

Growth of Labor Productivity

- Labor productivity is the quantity of real GDP produced by an hour of labor
- It is calculated by dividing real GDP by aggregate labor hours
- For example, if real GDP is \$13 trillion and aggregate hours are 200 billion, labor productivity is \$65 per hour
- When labor productivity grows, real GDP per person grows and brings a rising standard of living

Effects of an Increase in Labor Productivity



- If labor productivity increases, the quantity of real GDP that any given quantity of labor can produce increases
- If labor is more productive, the demand for labor also increases
- With an increase in the demand for labor (demand for labor curve shifts rightward) and no change in the supply of labor, the real wage rate rises and equilibrium quantity of labor also increases
- So an increase in labor productivity increases potential GDP for two reasons: Labor is more productive and more labor is employed

Why Labor Productivity Grows – Physical Capital Growth

- As the amount of capital per worker increases, labor productivity also increases
- The accumulation of capital on farms, in textile factories, in iron foundries and steel mills, in coal mines, on building sites, in chemical plants, in auto plants, in banks and insurance companies, and in shopping malls has added incredibly to the labor productivity of economies around the world

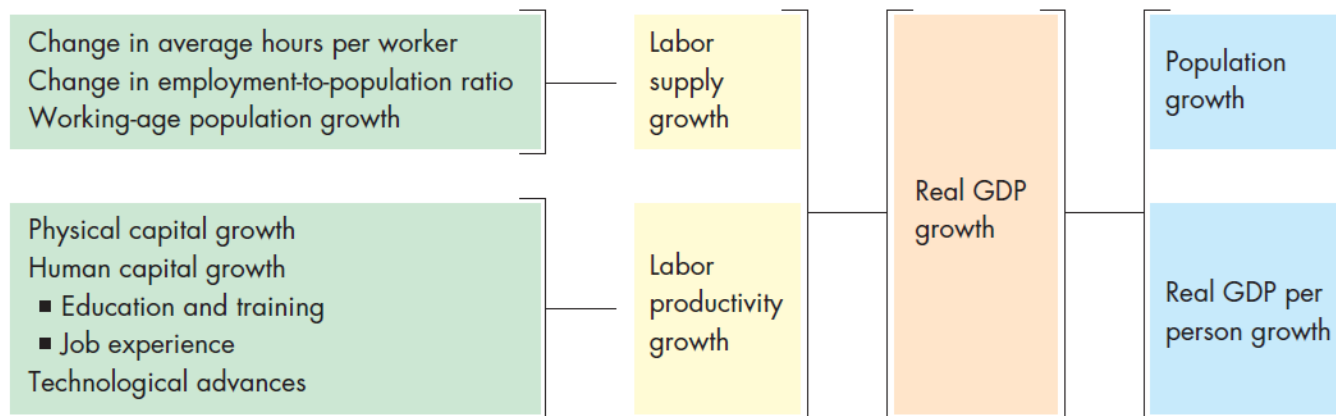
Why Labor Productivity Grows – Human Capital Growth

- Human capital—the accumulated skill and knowledge of human beings—is the fundamental source of labor productivity growth
- What are the examples you can think of today?

Why Labor Productivity Grows – Technological Advances

- The accumulation of physical capital and human capital have made a large contribution to labor productivity growth
- But technological change—the discovery and the application of new technologies—has made an even greater contribution

The Sources of Economic Growth



Labor supply growth and labor productivity growth combine to determine real GDP growth. Real GDP per person growth depends on real GDP growth and population growth.

Growth Theories (Self study)

Growth Theories (Self study)

Three main theories of economic growth:

- Classical growth theory
- Neoclassical growth theory
- New growth theory

Classical Growth Theory

- Classical growth theory is the view that the growth of real GDP per person is temporary and that when it rises above the subsistence level, a population explosion eventually brings it back to the subsistence level
- Modern-Day Malthusians believe
 - resource limits and population growth would lead to a decline in real GDP per person and world would return to a primitive standard of living
 - environmental issues like global warming and climate change as the reasons for real GDP per person to decrease eventually

Neoclassical Growth Theory

- Neoclassical growth theory is the proposition that real GDP per person grows because technological change induces saving and investment that make capital per hour of labor grow
- Growth ends if technological change stops because of diminishing marginal returns to both labor and capital

The Neoclassical Theory of Population Growth

- As women's wage rates increase and their job opportunities expand, the opportunity cost of having children increases. Faced with a higher opportunity cost, families choose to have fewer children and the birth rate falls
- Technological advances that bring higher incomes also brings advances in health care that extends lives. So as incomes increase, both the birth rate and the death rate decrease
- These opposing forces offset each other and result in a slowly rising population

Technological Change and Diminishing Returns

- Technological advances bring new profit opportunities — Businesses expand, and new businesses are created to exploit the newly available profitable technologies
- The economy enjoys new levels of prosperity and growth
- But will the prosperity last? And will the growth last?
- Neoclassical growth theory says that the prosperity will last but the growth will not last unless technology keeps advancing

A Problem with Neoclassical Growth Theory

- Real GDP growth rates and income levels per person around the world will converge
- But according to the real world data, it does not appear to be imminent for all countries

New Growth Theory

- The growth rate depends only on people's incentives and ability to innovate
- Profit is the spur to technological change
- Inventors can maintain a profit for several years by taking out a patent or a copyright, but eventually, a new discovery is copied, and profits disappear. So more research and development is undertaken in the hope of creating a new burst of profitable investment and growth
- Two facts about discoveries and technological knowledge play a key role in the new growth theory:
 - Discoveries are (at least eventually) a public capital good
 - Knowledge is a capital that is not subject to diminishing marginal returns

A Perpetual Motion Economy

- New growth theory sees the economy as a perpetual motion machine
- No matter how rich we become, our wants exceed our ability to satisfy them
- In the pursuit of a higher standard of living, human societies have developed incentive systems—markets, property rights, and money—that enable people to profit from innovation
- Wants and incentives create innovation, new and better products, and a yet higher standard of living

New Growth Theory Versus Malthusian Theory

- To a Malthusian, population growth is part of the problem
- To a new growth theorist, population growth is part of the solution
- People are the ultimate economic resource — a larger population brings forth more wants, but it also brings a greater amount of scientific discovery and technological advance
- So rather than being the source of falling real GDP per person, population growth generates faster labor productivity growth and rising real GDP per person
- Resources are limited, but the human imagination and ability to increase productivity are unlimited

Diagrams illustrating the effects on real wage, labor hours & real GDP are examinable